Dinesh Sheelam

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SUMMARY

Data Analyst with 3.5 years of experience in requirements gathering, data quality assurance, data cleaning, data modeling and building robust ETL processes and data warehouse solutions. Skilled in designing and monitoring key performance indicators (KPIs), delivering accurate reporting, and supporting marketing operations with actionable insights. Proven ability to translate industry trends into data driven strategies, automate workflows, and resolve technical issues through effective customer support. Currently seeking to transition into AI/ML focused data engineering roles to develop scalable data pipelines and predictive analytics solutions.

TECHNICAL SKILLS

- **Languages**: Python, SQL
- **Data Processing & ETL**: Pandas, NumPy, PySpark, Data Wrangling, Data Cleaning, Data Quality Assurance, Feature Engineering, **ETL Processes**, Batch Processing Pipelines
- **Machine Learning & Deep Learning**: scikit-learn, PyTorch, MONAI, EfficientNet, Faster R-CNN, Cross-Validation, Hyperparameter Tuning, Model Evaluation
- **Data Visualization**: Power BI, Tableau, Matplotlib, Seaborn, MS Excel, Streamlit, KPI Dashboards, Reporting Automation
- **Databases & Cloud Storage**: MySQL, PostgreSQL, Microsoft SQL Server, Snowflake, Google Cloud Storage, AWS, Data Warehouse Design
- **Business & Operations:** Requirements Gathering, Customer Support, Marketing Operations, Agile Methodologies
- **Tools & Frameworks**: Jupyter Notebook, Anaconda, Google Vertex AI, ML Pipelines, Git & GitHub, VS Code
- **Operating Systems**: Windows, Linux (Ubuntu), MacOS

WORK EXPERIENCE

CEO Capital Traders | Pune, India — ***Aug 2020 – Jul 2023***

Data Analyst

- **Designed and deployed** end-to-end data products by transforming raw transaction data into clean, scalable data models using tools like **Snowflake**, advanced **ETL processes**, and SQL pipelines powering automated Tableau dashboards that reduced identified risks by **20%**.
- **Orchestrated** requirements gathering sessions with cross functional teams and leadership, aligning business objectives with actionable **KPIs** and robust risk metrics to enhance fraud detection strategies.
- **Constructed** standardized workflows and risk analysis documentation, improving assessment speed by **20%** and ensuring high **data quality** across dynamic transaction datasets.
- **Engineered** and maintained an integrated **data warehouse** on Snowflake, leveraging best practices in **data cleaning**, schema design, and performance optimization for rapid querying and reporting.
- **Synthesized** complex fraud detection outputs into clear, business-ready insights, delivering strategic recommendations that cut fraud response time by **30%** and improved decision-making.
- **Championed** a culture of data integrity by conducting frequent audits, validating metrics, and leading root cause analyses (RCAs) enhancing client trust and increasing issue resolution speed by **20%**.
- **Facilitated** timely, high touch **customer support** for internal and external stakeholders by addressing escalations, communicating technical resolutions, and ensuring continuous data pipeline uptime.
- **Monitored** emerging **industry trends** and market conditions to proactively refine risk models and update key metrics, aligning analytics deliverables with evolving compliance standards.

KEKA Technologies | Hyderabad, India — ***Aug 2019 – Dec 2019***

Data and Business Analyst Intern

- **Conducted** end to end **requirements gathering** and translated user stories into technical solutions for payroll automation projects, ensuring alignment with defined **KPIs** and stakeholder goals.
- **Developed** robust SQL based databases and optimized data pipelines, guaranteeing high **data quality** and accelerating actionable sales intelligence by **15%**.
- **Analyzed** large-scale HR and payroll data sets using Python (NumPy, Pandas, scikit learn), implementing advanced **data cleaning** and feature engineering that improved business insights by **10%**.

- **Produced** dynamic Power BI dashboards with custom DAX measures and Power Query transformations, delivering clear **reporting** that empowered pricing optimizations and drove a **12%** revenue increase.
- **Collaborated** with product managers and marketing teams to align **marketing operations** with data-driven strategies, completing 8 major deliverables on time and within scope.
- **Evaluated** ad hoc data requests and audit requirements to verify report accuracy, ensuring data relevance and usability for leadership decision-making.

PROJECTS

RSNA 2024 Lumbar Spine Degenerative Classification | Southeast Missouri State University ***year*: 2024**

Python, PyTorch, MONAI, EfficientNet-B0, Albumentations, Google Vertex AI

- **Developed** an end-to-end lumbar spine disease classification model using PyTorch and MONAI, boosting test accuracy from ~75% (baseline CNN) to **80.9%** and reducing test loss to **0.5847** by integrating an EfficientNet-B0 backbone (timm).
- **Engineered** a two-stage detection-classification pipeline: disc-level detection with Faster R-CNN ResNet-50 FPN (precision ~90% at IoU 0.5, recall ~82%), followed by multi-class severity classification, significantly improving per-class balance.
- **Executed** 5-fold cross validation and advanced data augmentation (Albumentations) on **25,000+ DICOM spine slices**, achieving consistent, balanced probability outputs for canal stenosis, foraminal narrowing, and subarticular stenosis.
- **Optimized** training and inference workflows by migrating from Kaggle notebooks to **Google Vertex AI**, leveraging GPU acceleration and cloud cost management to handle **8,593 MRI series** (~2,697 patients).
- **Delivered** a robust, competition-ready inference pipeline generating probabilistic outputs for **25 degenerative spine conditions**, fully compliant with RSNA challenge standards and designed for real-world clinical workflows.

Investor Segmentation for Personalized Investment Strategies| CEO Capital Traders ***year*: 2022- 2023**

Python, pandas, scikit-learn, SQL, K-Means, PCA, matplotlib, Streamlit

- **Engineered** a behavioral clustering model to segment **30,000+ investors** based on **trading frequency**, **investment size**, **asset diversity**, and **risk profiles**.
- **Cleaned** and **merged** multi-source **customer**, **transaction**, and **asset data** using **Python** and **SQL**, ensuring high **data integrity** for analysis.
- **Applied K-Means clustering** and **PCA** to identify **4 distinct investor groups**, enabling **targeted marketing** and **personalized product recommendations**.
- **Visualized clusters** and segment traits with clear plots using **matplotlib**, improving stakeholder understanding and driving a **22% increase in customer engagement**.
- **Documented** a reusable **end to end analytics pipeline**, showcasing strong **data wrangling**, **feature engineering**, and **unsupervised learning** skills.

EDUCATION

South East Missouri State University, Missouri, MO. ***Graduated: May 2025***

Master of Science in Applied Computer Science, GPA 3.8/4.0

Relevant Coursework: Data Analysis, Distributed Computing, Soft Computing Techniques, Cyber Security, Data Mining, AI/ML.

Lovely Professional University, Punjab, India. ***Graduated: August 2020***

Bachelor of Engineering in Computer Science, GPA 7.1/10.0

Relevant Coursework: Probability, Statistics, Cloud Computing, Python Programming, Software Engineering, Data Analytics